MINITEC · AI STRATEGY

AI That Eliminates Wasted Engineering Hours

A proposal for deploying AI to solve MiniTec's two biggest operational bottlenecks.

Module: AI in Business

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[EXECUTIVE SUMMARY]

Here Is the Problem We Are Solving

MiniTec's engineers lose thousands of hours yearly to two fixable bottlenecks. AI solves both.

PROBLEM 1 · SEARCH GAP

Engineers recreate parts that already exist

Autodesk Vault's keyword-only search cannot interpret natural language. Engineers can't find existing models — so they redraw them, creating duplicates and wasting time.

60%

of parts may be duplicates

[1] Aberdeen Group, 2022

1,251 hrs

wasted / engineer / year

[2] McKinsey, 2021

✅ **AI Solution: Text-to-CAD Matchmaker — semantic search in plain English**

PROBLEM 2 · DOC BOTTLENECK

Creating manuals takes 80 hours of copy-paste

Writing one operating manual requires manually gathering data from CAD, ERP, BOM, and previous manuals. 80% of content is repetitive across product variants.

80 hrs

per manual (average)

[3] Internal MiniTec est.

33%

of engineer time is non-value-added

[2] McKinsey, 2021

✅ **AI Solutions: Auto Manual Generator + Watch & Write Video AI**

[STATUS QUO]

The Industry Has Moved — MiniTec Has Not

Competitors already deploy AI for search and documentation. We are behind.

Internal — Highly Manual Processes

- Autodesk Vault: keyword-only, no semantic understanding
- Manuals: 100% manual copy-paste across 4+ systems
- 80% of doc content is identical across product variants
- No single source of truth — updates lag weeks behind changes

[3] Internal MiniTec Process Audit

External — AI Adoption Is Here Now

- CADENAS PARTsolutions — AI shape & semantic CAD search
- Dassault EXALEAD — Multi-CAD deduplication platform
- Cortona3D RapidAuthor — Auto 3D manual generation from CAD
- LLM / RAG pipelines — Cut SOP drafting time by 60–75%

[4] Gartner AI in Manufacturing 2023 [5] CADENAS Product Docs

[STRATEGY & IDEATION]

How We Selected Our 3 AI Use Cases

From 10+ brainstormed ideas, we applied a structured AI relevance filter and MoSCoW analysis.

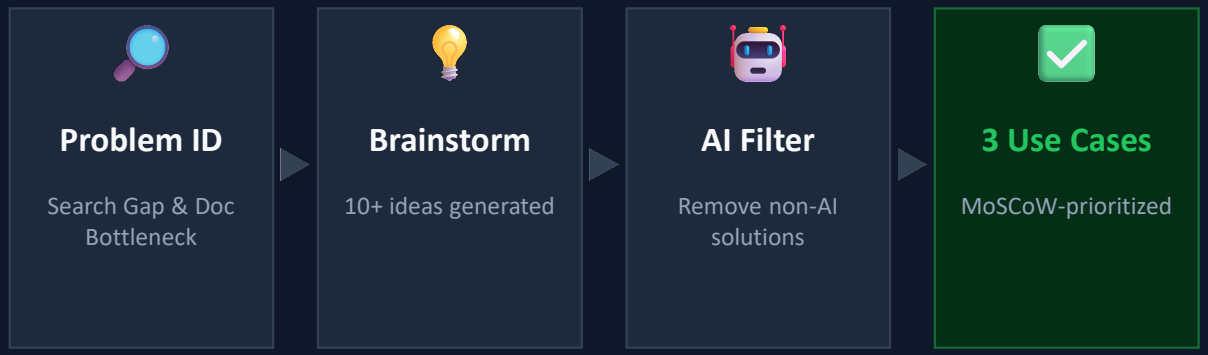
Overarching Strategy

Our strategy follows two mandatory principles:

1. Augmentative — AI assists, never replaces engineers.

2. Integrated — All AI plugs into existing Vault & ERP infrastructure.

This avoids rip-and-replace risk while delivering immediate ROI on existing systems.



MUST HAVE

Text-to-CAD Matchmaker

SHOULD HAVE

Auto Manual Generator

COULD HAVE

Watch & Write Video AI

[USE CASE 1 · SEARCH GAP]

Let Engineers Find Existing Parts in Plain English

Deploys NLP + Vector Search directly into Autodesk Vault to eliminate duplicate CAD creation.

How It Works

1 NLP Parser

Reads requirement text and extracts key parameters from plain English.

2 Vector Embedding Index

All Vault models indexed by semantic meaning — not just filename.

3 Ranked Results + Alert

Shows top matches with confidence score. Warns before saving a duplicate.

✅ **Type what the client asked for — the AI finds the part, even if it was saved under a completely different name.**

Use Case Metadata

Strategic Goal:

Reduce duplicate CAD parts by 90%

Target Users:

Design Engineers, Procurement

Key Tech:

NLP, Vector DB, Vault API

Timeline:

6–9 months (Phase 1)

SOURCES

[1] Aberdeen Group (2022) — Engineering duplicate rate study
[5] CADENAS PARTsolutions — AI Semantic CAD Search

[USE CASE 2 · DOC BOTTLENECK]

Cut Operating Manual Creation from 80 Hours to 2

An LLM that drafts manuals automatically from CAD, BOM, and ERP data — engineers just review.

How It Works

1 Data Pull (RAG)

AI retrieves specs from Vault, BOM, and ERP. No manual data-gathering required.

2 LLM Draft Generation

A fine-tuned language model writes the full manual draft using pre-approved templates.

3 Human Review & Approve

Engineer edits and signs off. AI is the drafter — the engineer is the editor.

⚠ **Safety: All AI-generated manuals require mandatory human sign-off. Compliant with CE/ISO.**

Use Case Metadata

Strategic Goal:

Cut doc time by 80–90%

Target Users:

Tech Writers, Product Managers

Key Tech:

LLM (RAG), Vault + ERP APIs

Timeline:

6–10 months (Phase 2)

SOURCES

[6] Cortona3D — Automated Technical Doc Generation
[7] McKinsey — LLMs in Industrial Documentation (2023)

[USE CASE 3 · DOC BOTTLENECK]

Let Technicians Record Assembly Guides — Not Type Them

Computer Vision AI converts technician recordings into step-by-step manuals automatically.

How It Works

1 Record (Smart Glasses / Phone)

Technician performs the assembly task while narrating out loud. No preparation needed.

2 AI Analyzes Video + Speech

Computer Vision identifies each step. Speech-to-text captures the narration.

3 Auto-Generate Written Guide

Outputs a formatted step-by-step document with screenshots. Zero typing required.

✅ **Innovation Leap: Moves documentation from the office to the factory floor.**

Use Case Metadata

Strategic Goal:

Zero-typing assembly documentation

Target Users:

Assembly Techs, Floor Supervisors

Key Tech:

Computer Vision, Speech-to-Text AI

Timeline:

12–18 months (Phase 3)

SOURCES

[8] PTC Vuforia — Video-based Work Instructions
[9] Google Vision AI — Action Recognition (2023)

[EVALUATION & PRIORITIZATION]

Rank Our 3 Use Cases by Impact and Feasibility

Star rating out of 5 per criterion. Weighted total score used to set rollout order.

Criteria (Weight)		UC1: Matchmaker	UC2: Auto Manuals	UC3: Video AI
Strategic Impact	30%	■■■■■	■■■■■	■■■■□
Implementation Feasibility	25%	■■■■□	■■■□□	■■□□□
Time to Value	25%	■■■■□	■■■■□	■■□□□
AI Innovation	20%	■■■■□	■■■■□	■■■■■
TOTAL SCORE		**4.35 / 5**	**3.90 / 5**	**3.05 / 5**
		#1 PRIORITY	#2 PRIORITY	#3 LONG-TERM

[FINANCIAL EVALUATION]

All 3 Use Cases Pay for Themselves Within 2 Years

Estimated investment vs. measurable returns per use case.

UC1: Text-to-CAD Matchmaker

~12 mo

PAYBACK PERIOD

INVESTMENTS

Dev & Integration	~€30k
Cloud Compute /yr	~€12k

RETURNS

Eng. hrs saved /yr	1,100 hrs
Duplicate reduction	-90%

[1] Aberdeen Group 2022 [2] McKinsey 2021

UC2: Auto Manual Generator

~15 mo

PAYBACK PERIOD

INVESTMENTS

Dev & LLM fine-tune	~€45k
LLM API tokens /yr	~€8k

RETURNS

Doc time saved	80h → 2h
Content reuse	+800%

[6] Cortona3D [7] McKinsey LLM Report 2023

UC3: Watch & Write Video AI

~24 mo

PAYBACK PERIOD

INVESTMENTS

Hardware + Dev	~€80k
AI Compute /yr	~€15k

RETURNS

Assembly doc time	Near Zero
Error reduction	~70%

[8] PTC Vuforia [9] Google Vision AI

[RISK MANAGEMENT]

We Have Mapped and Mitigated Every Key Risk

Risks categorized by type and severity. Each paired with a concrete mitigation strategy.

⚠ HIGH — LEGAL

IP Exposure

Proprietary CAD designs uploaded to public AI could expose MiniTec's trade secrets.

✅ Deploy all AI on-premise or private cloud. No data leaves the organization.

⚠ HIGH — SAFETY

AI Hallucination in Safety Manuals

LLM generates incorrect safety steps, risking user injury or CE non-compliance.

✅ Mandatory human engineer sign-off. Full audit trail maintained.

MED — ETHICAL

Job Displacement Fear

Tech writers and engineers may resist AI, fearing their roles will be eliminated.

✅ AI positioned as augmentation. Upskilling programs. Roles shift to higher-value tasks.

MED — TECHNICAL

Garbage In, Garbage Out

Inconsistent Vault metadata will cause poor AI search results if data is not cleaned.

✅ Phase 1 mandates full Vault metadata audit before AI deployment.

✓ LOW — SUSTAINABLE

Energy Consumption

Running AI inference at scale requires significant compute power.

✅ Use green-certified cloud providers. Optimize model size. Run batch inference.

✓ LOW — TECHNICAL

Integration Complexity

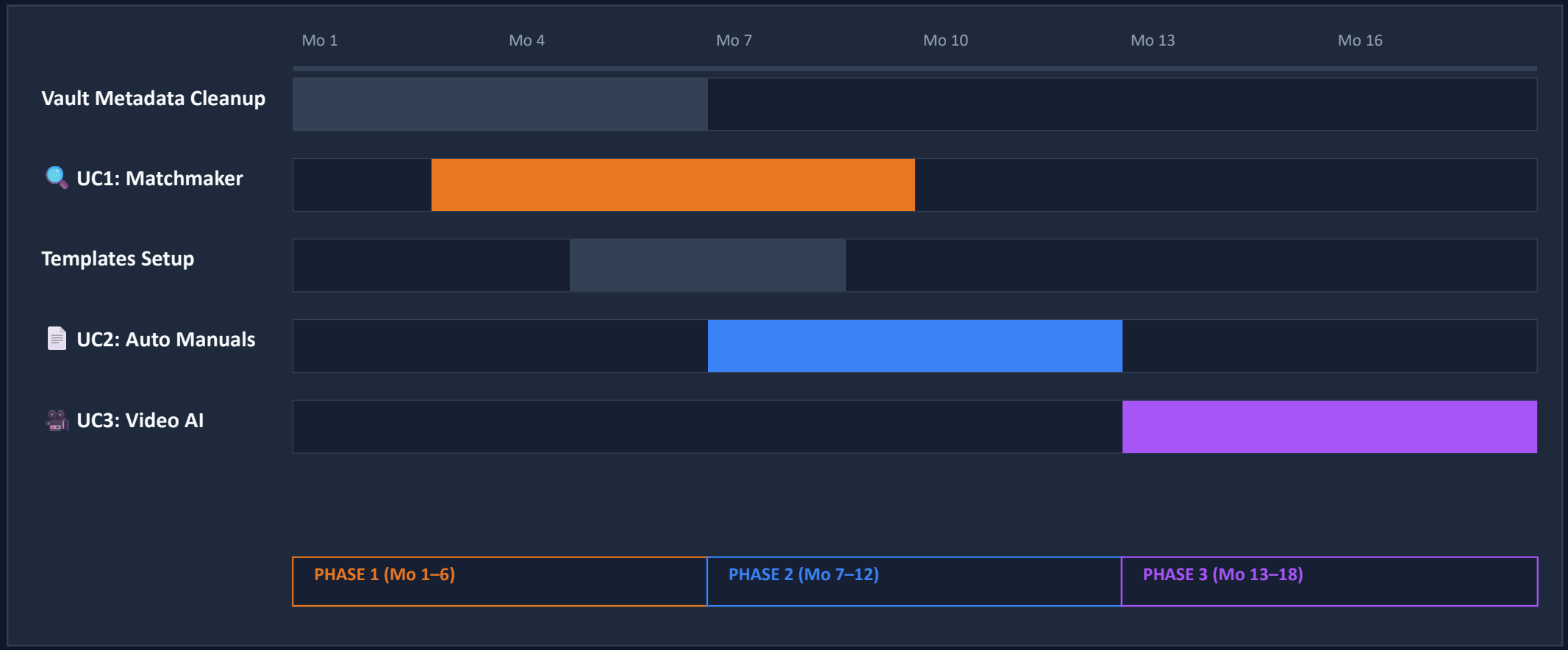
Connecting AI to Vault and ERP may uncover undocumented APIs or legacy issues.

✅ 4-week PoC in Phase 1 validates integration before full rollout.

[IMPLEMENTATION]

Deploy in 3 Phases Over 18 Months

A realistic, risk-reduced rollout — starting with quick wins and building to full AI coverage.



[CONCLUSION]

"The Art of Simplicity — Let AI handle the repetitive, so people can handle the complex."

Next Step: Approve Phase 1 & start a 4-week Proof of Concept

-90%	80h → 2h	+800%	~12 mo
Duplicate Parts	Manual Creation	Content Reuse	First Payback

Sources: [1] Aberdeen Group 2022 [2] McKinsey 2021 [3] Internal Estimate [4] Gartner 2023 [5] CADENAS Docs [6] Cortona3D [7] McKinsey LLM 2023 [8] PTC Vuforia [9] Google Vision AI

References & Sources

[1] Aberdeen Group (2022) — Engineering Efficiency and the CAD Duplication Problem

[2] McKinsey & Company (2021) — The Economic Potential of Generative AI

[3] MiniTec Internal Estimate — Manual Documentation Process Audit

[4] Gartner Research (2023) — AI in Manufacturing: Adoption Trends

[5] CADENAS PARTsolutions — AI-Based Geometric & Semantic Part Search

[6] Cortona3D RapidAuthor — Automated Technical Documentation from CAD

[7] McKinsey & Company (2023) — Generative AI in Industrial Documentation

[8] PTC Vuforia Instruct — Video-Based Work Instructions Platform

[9] Google Cloud Vision AI — Action Recognition and Video Intelligence